

A Regime-Aware Model for Loan-Level Probability of Default

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Literature Review

The literature on credit-risk modelling is unified by a common tension: default risk is observed at the borrower or loan level, but it evolves through time with the macro-financial environment. Borrowers differ in leverage, affordability, credit history, and loan terms; however, the risk carried by those characteristics is not necessarily invariant across economic states. A highly leveraged mortgage, for example, is more vulnerable when house prices fall or refinancing conditions tighten than when collateral values and liquidity are improving. This review therefore traces the literature from static borrower-level scorecards to models that incorporate macroeconomic covariates, latent regimes, and probability calibration. The objective is to position the thesis as a structured extension of interpretable loan-level probability-of-default (PD) modelling rather than as a general claim that regime-switching methods are new to credit risk.

1.1 Loan-Level PD Modeling

Credit risk modelling has long been organised around the estimation of default likelihood. In structural models, default is linked to the value of the obligor’s assets relative to promised debt payments (Merton, 1974). In reduced-form models, default is treated as a stochastic event with an arrival intensity that may depend on observable covariates, latent economic conditions, or both (Jarrow and Turnbull, 1995; Lando, 1998; Duffie and Singleton, 2003). Retail credit and mortgage applications typically operationalise this idea at the account or loan level, where the modelling target is a conditional PD for a particular borrower in a particular observation window. If Y_i denotes the default outcome for borrower or loan i , and x_i denotes observed borrower and loan characteristics, then a standard scoring model estimates

$$\Pr(Y_i = 1 \mid x_i) = F(\eta_i), \quad \eta_i = \beta_0 + x_i^\top \beta, \quad (1.1)$$

where $F(\cdot)$ is a link function. Earlier empirical credit-risk scorecards used discriminant-analysis methods to classify financial distress (Altman, 1968); later consumer-credit scoring work made logistic regression a central benchmark. The logistic link, $F(a) = (1 + \exp(-a))^{-1}$, remains common because it maps linear scores into probabilities in $[0, 1]$, produces coefficients with an odds-ratio interpretation, and is relatively easy to validate and communicate (Hand and Henley, 1997; Thomas, 2000; Costa e Silva et al., 2020). The persistence of logistic scorecards is not due to superior flexibility, but to their transparency, statistical stability, and compatibility with model-risk governance.

In mortgage credit risk, the relevant predictors usually include leverage, affordability,

credit history, loan product, occupancy, property type, and seasoning. The economic mechanisms are well established. Higher loan-to-value ratios increase vulnerability to negative equity; higher debt-to-income ratios reduce repayment capacity; and credit scores, delinquency history, and loan age proxy for borrower quality, payment discipline, and the changing risk profile of a mortgage over its life. Mortgage default is also connected to the exercise of embedded borrower options, because default and prepayment decisions depend on collateral values, refinancing incentives, and household circumstances (Deng et al., 2000; Campbell and Cocco, 2015). Empirical hazard models likewise show that default timing is shaped by both underwriting conditions and subsequent shocks, particularly changes in house prices (Palmer, 2023).

The principal advantage of loan-level PD models is that they preserve the unit at which default is observed. Portfolio default rates identify broad cycles, but they collapse the heterogeneity needed for underwriting, monitoring, and risk segmentation. Loan-level models allow the same calendar month to generate different PDs for different borrowers and support portfolio-loss analysis through aggregation of individual risk estimates. Recent mortgage credit-risk work using large U.S. loan-level datasets continues this tradition, while increasingly adding nonlinear, spatial, or machine-learning components to capture heterogeneity that conventional linear models may miss (Kundig and Sigrist, 2025).

The limitation of the simplest formulation in Equation 1.1 is that the coefficient vector β is treated as stable. The model assumes that the marginal effect of each borrower feature is constant across calendar time. Static models may be defensible over short and economically stable samples, but they become less satisfactory when inflation, interest rates, unemployment, house prices, and credit conditions shift materially. A borrower-only model may still rank loans well within a month, yet produce poorly calibrated PDs across months if the same score is mapped to the same probability under very different macroeconomic conditions. This distinction between discrimination and probability accuracy is central to the present thesis.

1.2 Macro-Augmented Credit Risk Models

A natural response to time variation is to augment borrower-level models with macroeconomic variables. Dynamic default models and hazard specifications are especially relevant because they model the timing of default rather than only a cross-sectional default indicator (Shumway, 2001). In a macro-augmented loan-level model, the PD is written as a function of both borrower or loan predictors $z_{i,t}$ and macroeconomic variables m_t :

$$\Pr(Y_{i,t} = 1 \mid z_{i,t}, m_t) = F(\beta_0 + z_{i,t}^\top \beta + m_t^\top \gamma). \quad (1.2)$$

This structure creates a direct channel from the economy to default probabilities. It is particularly relevant in stress testing, where adverse macroeconomic scenarios must be translated into changes in PDs and portfolio losses.

Bellotti and Crook (2009) show that macroeconomic variables such as interest rates and unemployment can be incorporated into credit scoring through survival analysis, where time-varying covariates are represented more naturally than in a single-period scorecard. Their

contribution is to connect credit scoring with default timing. Bellotti and Crook (2013) extend this logic to retail credit stress testing, using discrete survival analysis with macroeconomic factors to estimate default rates under downturn scenarios. The broader stress-testing literature similarly treats macroeconomic variables as the bridge between credit-risk models and forward-looking risk management (Simons and Rolwes, 2009).

Macro-augmented models clarify that default probabilities are not determined only by borrower characteristics. Unemployment affects income stability, interest rates affect debt-service burdens and refinancing options, inflation affects real disposable income, and house prices affect borrower equity and recovery expectations. These channels are especially relevant for mortgages because the collateral value is itself cyclical. A macro-augmented model can therefore improve level calibration by shifting average PDs upward or downward as economic conditions deteriorate or improve.

The standard macro-augmented formulation also has an important restriction. Equation 1.2 treats macroeconomic effects as additive and coefficient-stable unless interactions or nonlinear terms are introduced. The macro vector shifts the intercept or adds fixed macro effects, but it does not by itself allow borrower sensitivities to change. This matters because economic stress may not raise all default probabilities by a common amount. It may increase the effect of high debt-to-income ratios, weak credit history, or high leverage. In such cases, the macro environment changes not only the level of default risk but also the slope of borrower risk.

This distinction motivates interaction-based and state-dependent approaches. If borrower effects vary by macro condition, the relevant object is closer to

$$\Pr(Y_{i,t} = 1 \mid z_{i,t}, m_t) = F \left(\beta_0 + z_{i,t}^\top \beta + m_t^\top \gamma + \sum_{j=1}^p \sum_{\ell=1}^q \theta_{j\ell} z_{i,j,t} m_{\ell,t} \right). \quad (1.3)$$

This specification is more flexible, but it can become unstable when many borrower and macro variables are interacted, particularly in rare-event default data. It also treats the observed macro variables themselves as state descriptors, even though stress is often inferred from a combination of variables rather than observed directly in one series. This limitation motivates the transition from observed macro covariates to latent macro states.

1.3 Regime-Switching Approaches

Regime-switching models formalise the idea that economic time series may follow different probability laws across unobserved states. Rather than imposing one parameter set on all periods, a Markov-switching model introduces a latent state process that changes the parameters of the observation equation. Hamilton (1989) applied this idea to economic time series by allowing output growth to switch across latent business-cycle states. Rabiner (1989) provides the general hidden Markov model (HMM) framework in which the state process is unobserved and inferred from the observed sequence.

For a macroeconomic observation vector M_t , a basic finite-state HMM assumes

$$\Pr(R_t = k \mid R_{t-1} = j, R_{t-2}, \dots, M_{1:t-1}) = \Pr(R_t = k \mid R_{t-1} = j) = p_{jk}, \quad (1.4)$$

where R_t is the latent regime and $P = [p_{jk}]$ is the transition matrix. Conditional on the regime, the observed macro vector follows a state-dependent distribution. Under a Gaussian emission model,

$$M_t \mid R_t = k \sim \mathcal{N}(\mu_k, \Sigma_k). \quad (1.5)$$

The model therefore describes both persistence and switching. Stress is not treated as a deterministic break date selected after the fact; it is a probabilistic state inferred from the data and governed by a transition law.

Credit risk applications of hidden-state models show why this structure is attractive. Yu et al. (2018) model credit risk with hidden Markov default intensity, where hidden economic states influence default intensities and the parameters are estimated using the expectation-maximisation algorithm. Their empirical results indicate that extracted hidden states can align with observed periods of economic stress and can be useful for credit-risk measurement. Milidonis and Chisholm (2024) extend structural default-risk modelling by allowing default-relevant asset dynamics to switch across regimes. Their simulations and empirical analysis show that a regime-switching structural model can respond more quickly to sudden changes in default risk than a constant-parameter Merton-type benchmark. Together, these studies support the view that default risk is often poorly represented by a single unconditional parameter set.

Regime-switching models are closely related to, but distinct from, macro-augmented scorecards. Macro-augmented models condition directly on observed variables such as unemployment and interest rates. HMM-based approaches use those variables to infer a lower-dimensional latent state, often interpreted as expansion or stress. The latent state is useful because stress is not always identified by one macro series. One period may involve rising rates, moderate unemployment, and high inflation; another may involve labour-market weakness and lower inflation. An HMM summarises the joint behaviour of the macro vector into a state probability:

$$\hat{p}_t^{\text{Stress}} = \Pr(R_t = \text{Stress} \mid M_{1:t}, \hat{\Theta}). \quad (1.6)$$

This filtered probability is a soft measure of macro stress intensity. It avoids forcing each month into a hard label when evidence is mixed and can be used as an input to downstream PD modelling.

Recent regime-aware credit forecasting work has moved in this direction. A closely related recent preprint by Brookfield et al. (2026) treats macro sensitivity and regime behaviour as central to consumer-credit forecasting. Because the study is a preprint rather than a peer-reviewed article, it is best used as contemporary motivation rather than as settled evidence. Its relevance for this thesis is that it combines regime detection, consumer-credit prediction, and regime-specific calibration, thereby sharpening the empirical problem: non-regime models may perform adequately in average conditions but respond too slowly when the macroeconomic environment changes.

The present thesis is positioned differently. It does not attempt to replicate a gradient boosting architecture or compete with high-dimensional machine-learning systems on flexibility. Instead, it asks a narrower mathematical and empirical question: can a filtered macro stress probability, estimated from a two-state HMM, improve an interpretable mortgage PD model when introduced through borrower-feature interactions? This makes the proposed

framework closer to a structured extension of logistic credit scoring than to a black-box forecasting model.

This distinction is important for publication framing. The novelty is not the use of hidden states in credit risk; the literature already establishes that possibility. The contribution is the coupling of a latent macro-regime layer with a transparent loan-month logistic PD layer, with calibration treated as a primary outcome. The model is intended to be auditable: each borrower coefficient has a baseline interpretation, each interaction coefficient has a regime-sensitivity interpretation, and the macro-regime input is a probability derived from a separately specified stochastic process.

The literature can therefore be summarised as a sequence of modelling choices. Table 1.1 maps these choices to the contribution of the present thesis.

Table 1.1: Synthesis of the main literature streams supporting the thesis

Literature stream	Main contribution	Remaining issue addressed here
Loan-level PD models	Estimate borrower-specific default probabilities using interpretable borrower and loan characteristics.	Borrower effects are often treated as stable across macroeconomic conditions.
Macro-augmented models	Add observed economic variables to default models and stress-testing frameworks.	Macro effects are often additive and may not capture changing borrower sensitivities.
Regime-switching models	Infer persistent latent economic states and allow model dynamics to differ across regimes.	Many applications focus on market or portfolio credit risk rather than interpretable loan-level mortgage PD.
Calibration literature	Evaluates whether predicted PDs match observed default frequencies, not only whether rankings are accurate.	Regime-aware gains should be judged by probability quality over time and during stress.

1.4 Gap Addressed By This Thesis

The gap targeted by this thesis is practical and methodological. Existing loan-level PD models are interpretable, but many standard specifications are static. Macro-augmented models introduce economic conditions, but their baseline form imposes additive and time-invariant effects. Regime-switching credit models capture latent stress, but many applications are developed for corporate credit, market instruments, or portfolio-level forecasting rather than

transparent mortgage loan-level PD. Machine-learning mortgage models can capture nonlinearities and interactions, but their flexibility often reduces coefficient-level interpretability, which remains valuable in a mathematical thesis and in regulated credit-risk settings (Kundig and Sigrist, 2025).

The thesis therefore studies an intermediate framework. It keeps the logistic PD model as the borrower-level prediction engine, estimates macro stress through a Gaussian HMM, and allows the stress probability to interact with borrower characteristics. The resulting model tests whether latent macro stress changes borrower risk sensitivities:

$$\text{logit}(\pi_{i,t}) = \beta_0 + z_{i,t}^\top \beta + \gamma \hat{p}_t^{\text{Stress}} + \hat{p}_t^{\text{Stress}} z_{i,t}^\top \theta. \quad (1.7)$$

If the interaction vector θ is materially different from zero and improves out-of-sample calibration, the evidence would support the claim that macro regimes affect not only the average default level but also the mapping from borrower features to default probability.

This framing also explains the calibration-first evaluation strategy. In credit risk, a model that ranks loans well but produces inaccurate PD levels can still be unsuitable for provisioning, stress testing, pricing, and portfolio monitoring. The Brier score and its decomposition evaluate probabilistic accuracy (Brier, 1950; Murphy, 1973); reliability diagrams provide visual evidence on calibration across probability bins (Dimitriadis et al., 2020); and PD calibration literature emphasises that estimated default probabilities should align with observed default frequencies, not merely sort loans correctly (Tasche, 2013; Fonseca and Lopes, 2017). Precision-recall measures are also relevant because serious delinquency is a rare event, and class imbalance can make accuracy and some ranking summaries incomplete (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).

The literature therefore supports the research question: whether an HMM-derived macro stress probability can improve the calibration and temporal stability of an interpretable loan-level mortgage PD model relative to borrower-only and macro-augmented logistic baselines. The methodology chapter formalises this question in terms of data construction, latent regime inference, model estimation, and chronological out-of-sample evaluation.

Methodology

2.1 Data Sources

The empirical workflow combines two public data sources: Freddie Mac sample mortgage loan files and monthly macroeconomic series from the Federal Reserve Economic Data (FRED) database (Freddie Mac, 2026; Federal Reserve Bank of St. Louis, 2026). The mortgage files provide loan-level origination characteristics and monthly servicing performance records. The macroeconomic data provide the aggregate state variables used to infer the latent regime process.

Table 2.1: Data sources and empirical role

Source	Main variables	Role in the methodology
Freddie Mac sample origination files	Borrower and loan characteristics at origination, including credit quality, leverage, product type, and loan terms.	Define the cross-sectional risk profile of each mortgage.
Freddie Mac sample servicing files	Monthly performance status, delinquency indicators, remaining balance, current activity, and loan age.	Construct the loan-month panel and forward-looking default label.
FRED macroeconomic series	Inflation, Federal Funds rate, and unemployment rate at monthly frequency.	Estimate the hidden macro regime and produce filtered stress probabilities.

Let $i = 1, \dots, N$ index loans and let $t = 1, \dots, T$ index calendar months. The raw mortgage records are transformed into a panel in which each observation corresponds to a loan-month. The origination file supplies time-invariant or slowly varying underwriting characteristics, while the servicing file supplies monthly performance information. The macroeconomic variables are merged by calendar month so that all loans observed in month t receive the same macro vector.

The monthly macro vector is denoted by

$$M_t = \begin{bmatrix} \iota_t \\ r_t \\ u_t \end{bmatrix}, \quad t = 1, \dots, T, \quad (2.1)$$

where ι_t is the monthly inflation measure, r_t is the Federal Funds rate, and u_t is the unemployment rate. These variables are selected because they represent three major channels through which macroeconomic stress can affect mortgage repayment: purchasing-power pressure, financing-cost pressure, and labour-income risk.

Before estimating the hidden Markov model, the macro series are standardised to remove scale differences:

$$\widetilde{M}_{t,j} = \frac{M_{t,j} - \bar{M}_j}{s_j}, \quad j = 1, 2, 3, \quad (2.2)$$

where $\bar{M}_j$ and s_j are the mean and sample standard deviation of macro variable j estimated on the HMM training period. The same fitted standardisation constants are then applied to validation and test months. This prevents the transformation from using future test-period information and ensures that the Gaussian emission likelihood is not dominated by the variable with the largest numerical scale.

2.2 Unit Of Analysis

The unit of observation is one mortgage at one month. For loan i in month t , the full observation is written as

$$\mathcal{O}_{i,t} = \left(Y_{i,t}^{(12)}, X_{i,t}, M_t, \hat{p}_t^{\text{Stress}} \right), \quad (2.3)$$

where $Y_{i,t}^{(12)}$ is the 12-month serious-delinquency label, $X_{i,t}$ is the borrower and loan feature vector, M_t is the observed macro vector, and $\hat{p}_t^{\text{Stress}}$ is the filtered probability that month t is in the stress regime.

The borrower and loan feature vector is represented as

$$X_{i,t} = (X_{i,t,1}, X_{i,t,2}, \dots, X_{i,t,p})^\top \in \mathbb{R}^p. \quad (2.4)$$

The intercept is modelled separately in the logistic equations. The components of $X_{i,t}$ consist of origination variables and current performance variables observed at month t . The feature list is fixed before model estimation and excludes any variable measured after month t . The construction follows the information principle that only information available at or before month t may be used to predict the event over months $t + 1$ through $t + 12$.

The empirical panel is defined over the at-risk population. A loan-month is at risk when the loan is active and not already seriously delinquent at month t . Let $D_{i,t}$ denote the observed delinquency status at month t , and let $O_{i,t+h}$ be an indicator that delinquency status is observed for loan i in forward month $t + h$. The full-window indicator is

$$W_{i,t}^{(12)} = \prod_{h=1}^{12} O_{i,t+h}. \quad (2.5)$$

Let $\mathcal{A}_{i,t}$ denote the event that the loan is active and eligible for forward prediction. The modelling sample is restricted to

$$\mathcal{R} = \left\{ (i, t) : \mathcal{A}_{i,t} = 1, D_{i,t} < 90, W_{i,t}^{(12)} = 1 \right\}. \quad (2.6)$$

This restriction requires an observed delinquency status for every month in the 12-month look-ahead window. It prevents the model from learning from observations that are already in the target state and avoids using incomplete forward windows for label construction.

2.3 Primary Label

The primary response variable is a 12-month transition to serious delinquency, defined as reaching 90 or more days past due within the next 12 months. For each at-risk loan-month $(i, t) \in \mathcal{R}$, the label is

$$Y_{i,t}^{(12)} = \mathbf{1} \left\{ \max_{h=1,\dots,12} D_{i,t+h} \geq 90 \right\}. \quad (2.7)$$

This definition aligns the statistical target with a forward-looking PD concept: the question is not whether the loan is currently distressed, but whether it transitions into serious delinquency over a fixed prediction horizon.

The conditional probability of interest is

$$\pi_{i,t} = \Pr \left(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t \right), \quad (2.8)$$

where $\mathcal{F}_t$ denotes the information available at month t . In the borrower-only baseline, $\mathcal{F}_t$ contains loan-level information only. In the macro-augmented baseline, it also contains observed macro variables. In the regime-aware model, it contains the filtered stress probability inferred from the macro HMM.

The 90+ days-past-due threshold is used because serious delinquency is a standard operational default proxy in credit risk and is closely related to regulatory default definitions based on material arrears (Basel Committee on Banking Supervision, 2004; Board of Governors of the Federal Reserve System, 2006). The fixed 12-month horizon creates a consistent target across the panel and supports direct comparison of predicted probabilities with realised default frequencies.

2.4 Macro Regime Inference

The latent macro regime is modelled using a two-state Gaussian hidden Markov model. Let

$$R_t \in \{1, 2\} \quad (2.9)$$

denote the unobserved macroeconomic regime in month t . After estimation, the two states are interpreted economically and labelled as Expansion and Stress according to their fitted macro profiles.

The regime process is assumed to follow a first-order homogeneous Markov chain:

$$\Pr(R_t = k \mid R_{t-1} = j, R_{t-2}, \dots, \widetilde{M}_{1:t-1}) = \Pr(R_t = k \mid R_{t-1} = j) = p_{jk}, \quad (2.10)$$

for $j, k \in \{1, 2\}$. The transition matrix is

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}, \quad p_{jk} \geq 0, \quad \sum_{k=1}^2 p_{jk} = 1. \quad (2.11)$$

The diagonal entries p_{11} and p_{22} measure regime persistence. The off-diagonal entries measure transition probabilities between expansion and stress.

Conditional on the latent regime, the standardised macro vector follows a multivariate Gaussian distribution:

$$\widetilde{M}_t \mid R_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad k = 1, 2. \quad (2.12)$$

The complete HMM parameter vector is

$$\Theta = (\delta, P, \mu_1, \mu_2, \Sigma_1, \Sigma_2), \quad (2.13)$$

where $\delta = (\delta_1, \delta_2)$ is the initial regime distribution.

For an observed macro path $\widetilde{M}_{1:T}$ and a latent regime path $R_{1:T}$, the joint density is

$$f(\widetilde{M}_{1:T}, R_{1:T}; \Theta) = \delta_{R_1} \left[\prod_{t=2}^T p_{R_{t-1}, R_t} \right] \left[\prod_{t=1}^T \phi_3(\widetilde{M}_t; \mu_{R_t}, \Sigma_{R_t}) \right], \quad (2.14)$$

where $\phi_3(m; \mu, \Sigma)$ denotes the value of the three-dimensional Gaussian density at $m \in \mathbb{R}^3$ with mean vector μ and covariance matrix Σ :

$$\phi_3(m; \mu, \Sigma) = \frac{1}{(2\pi)^{3/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (m - \mu)^\top \Sigma^{-1} (m - \mu) \right\}, \quad (2.15)$$

with mean vector $\mu \in \mathbb{R}^3$ and positive definite covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$.

Because the regime path is unobserved, direct maximisation of the complete-data likelihood is not feasible. The model is estimated with the Baum-Welch algorithm, an expectation-maximisation procedure for HMMs (Baum et al., 1970; Bilmes, 1998). In the E-step at iteration q , the smoothed state probabilities are computed:

$$\gamma_t(k) = \Pr(R_t = k \mid \widetilde{M}_{1:T}; \Theta^{(q)}), \quad (2.16)$$

and the smoothed transition probabilities are

$$\xi_t(j, k) = \Pr(R_{t-1} = j, R_t = k \mid \widetilde{M}_{1:T}; \Theta^{(q)}), \quad t = 2, \dots, T. \quad (2.17)$$

In the M-step, the parameters are updated by weighted averages:

$$\delta_k^{(q+1)} = \gamma_1(k), \quad (2.18)$$

$$p_{jk}^{(q+1)} = \frac{\sum_{t=2}^T \xi_t(j, k)}{\sum_{t=2}^T \sum_{\ell=1}^2 \xi_t(j, \ell)}, \quad (2.19)$$

$$\mu_k^{(q+1)} = \frac{\sum_{t=1}^T \gamma_t(k) \widetilde{M}_t}{\sum_{t=1}^T \gamma_t(k)}, \quad (2.20)$$

$$\Sigma_k^{(q+1)} = \frac{\sum_{t=1}^T \gamma_t(k) (\widetilde{M}_t - \mu_k^{(q+1)}) (\widetilde{M}_t - \mu_k^{(q+1)})^\top}{\sum_{t=1}^T \gamma_t(k)}. \quad (2.21)$$

The E-step and M-step are iterated until the observed-data log-likelihood stabilises:

$$\ell(\Theta) = \log \left[\sum_{R_1=1}^2 \cdots \sum_{R_T=1}^2 f(\widetilde{M}_{1:T}, R_{1:T}; \Theta) \right]. \quad (2.22)$$

For downstream PD prediction, the relevant macro signal is not the smoothed probability in Equation 2.16. Smoothed probabilities condition on the full macro path $\widetilde{M}_{1:T}$ and therefore use future information. The PD model instead uses the filtered probability, computed from information available through month t :

$$\alpha_t(k) = f(\widetilde{M}_{1:t}, R_t = k; \hat{\Theta}), \quad (2.23)$$

with recursion

$$\alpha_1(k) = \hat{\delta}_k \phi_3(\widetilde{M}_1; \hat{\mu}_k, \hat{\Sigma}_k), \quad (2.24)$$

$$\alpha_t(k) = \phi_3(\widetilde{M}_t; \hat{\mu}_k, \hat{\Sigma}_k) \sum_{j=1}^2 \alpha_{t-1}(j) \hat{p}_{jk}, \quad t = 2, \dots, T. \quad (2.25)$$

$$\hat{p}_t(k) = \Pr(R_t = k \mid \widetilde{M}_{1:t}; \hat{\Theta}) = \frac{\alpha_t(k)}{\sum_{\ell=1}^2 \alpha_t(\ell)}. \quad (2.26)$$

The stress input used in the credit model is

$$\hat{p}_t^{\text{Stress}} = \hat{p}_t(k^*), \quad k^* = \arg \max_{k \in \{1,2\}} s(k), \quad (2.27)$$

where $s(k)$ is an economic stress score assigned from the fitted state profile. Since the three macro variables are oriented so that larger standardised values indicate greater stress pressure, the study uses

$$s(k) = \mu_{k,1} + \mu_{k,2} + \mu_{k,3}, \quad k = 1, 2. \quad (2.28)$$

Thus, the stress state is the fitted state with the larger combined standardised inflation, Federal Funds rate, and unemployment mean.

2.5 Model Comparison

The empirical design compares exactly three logistic specifications. The comparison is intentionally hierarchical: each model adds one source of information to the previous one while preserving a common loan-month sample, label definition, and chronological evaluation design.

The borrower-only baseline is

$$\pi_{i,t}^{(0)} = \Pr(Y_{i,t}^{(12)} = 1 \mid X_{i,t}) = \sigma(\beta_0 + X_{i,t}^\top \beta), \quad (2.29)$$

where $\sigma(a) = (1 + \exp(-a))^{-1}$. This model assumes that borrower and loan characteristics have fixed effects through time.

Table 2.2: Model hierarchy used in the empirical comparison

Model	Specification	Purpose
Borrower-only logit	Borrower and loan features $X_{i,t}$ only.	Establish the static loan-level benchmark.
Macro-augmented logit	Borrower features plus standardised macro variables $\widetilde{M}_t$.	Test whether observed macro variables improve PD estimation additively.
Regime-aware logit	Borrower features, filtered stress probability, and stress-by-borrower interactions.	Test whether latent stress changes borrower risk sensitivities.

The macro-augmented baseline is

$$\pi_{i,t}^{(M)} = \sigma \left(\beta_0 + X_{i,t}^\top \beta + \widetilde{M}_t^\top \gamma_M \right). \quad (2.30)$$

This model allows the macroeconomic environment to shift the PD, but it does not allow the effect of borrower variables to change with the macro state.

The proposed regime-aware model is

$$\pi_{i,t}^{(R)} = \sigma \left(\beta_0 + X_{i,t}^\top \beta + \gamma_R \hat{p}_t^{\text{Stress}} + \hat{p}_t^{\text{Stress}} X_{i,t}^\top \theta \right). \quad (2.31)$$

The coefficient vector β measures baseline borrower effects. The scalar γ_R measures the direct level effect of macro stress probability. The interaction vector θ measures how borrower sensitivities change as the economy moves toward the stress regime.

To see the interpretation of the interaction terms, consider a continuous borrower feature $X_{i,t,j}$. The marginal effect on the log-odds is

$$\frac{\partial}{\partial X_{i,t,j}} \text{logit} \left(\pi_{i,t}^{(R)} \right) = \beta_j + \hat{p}_t^{\text{Stress}} \theta_j. \quad (2.32)$$

When $\theta_j > 0$, the log-odds contribution of feature j becomes stronger in stress. When $\theta_j < 0$, the contribution weakens as stress probability rises. For binary or categorical indicators, the same expression is interpreted as a stress-dependent discrete log-odds difference rather than a derivative. This is the main mathematical distinction between the regime-aware model and the macro-augmented baseline.

The logistic coefficients are estimated by maximum likelihood. For a generic model with predicted probabilities $\hat{\pi}_{i,t}$, the Bernoulli log-likelihood over the training set $\mathcal{T}$ is

$$\ell = \sum_{(i,t) \in \mathcal{T}} \left[Y_{i,t}^{(12)} \log(\hat{\pi}_{i,t}) + (1 - Y_{i,t}^{(12)}) \log(1 - \hat{\pi}_{i,t}) \right]. \quad (2.33)$$

Because the regime-aware model contains a block of interaction terms, ridge regularisation is used to stabilise estimation:

$$\widehat{\Omega} = \arg \min_{\Omega} \left\{ -\ell(\Omega) + \lambda \|\theta\|_2^2 \right\}, \quad \lambda \geq 0, \quad (2.34)$$

where $\Omega = (\beta_0, \beta, \gamma_R, \theta)$ is the full coefficient vector for the regime-aware model. Penalising θ directly reflects the modelling goal: the interaction layer is allowed to improve the model, but unnecessary regime sensitivity is shrunk toward zero (le Cessie and van Houwelingen, 1992). The tuning parameter λ is selected on the chronological validation set by minimising the validation Brier score.

2.6 Evaluation Plan

All comparisons use chronological splits. This is necessary because the task is forward-looking: a credit model trained with future information would overstate performance. Let $\tau_1 < \tau_2$ denote calendar-month cutoffs. The full modelling sample is partitioned into training, validation, and test periods as

$$\begin{aligned}\mathcal{R}_{\text{train}} &= \{(i, t) \in \mathcal{R} : t \leq \tau_1\}, \\ \mathcal{R}_{\text{valid}} &= \{(i, t) \in \mathcal{R} : \tau_1 < t \leq \tau_2\}, \\ \mathcal{R}_{\text{test}} &= \{(i, t) \in \mathcal{R} : t > \tau_2\}.\end{aligned}\tag{2.35}$$

The HMM standardisation constants and HMM parameters are estimated using macro months up to τ_1 only. Filtered probabilities for validation and test months are then generated recursively with the fitted HMM parameters and macro observations observed up to each month. The downstream PD models are fit on $\mathcal{R}_{\text{train}}$, tuned on $\mathcal{R}_{\text{valid}}$, and evaluated once on $\mathcal{R}_{\text{test}}$.

The primary evaluation criterion is calibration. The Brier score is

$$BS = \frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} \left(\hat{\pi}_{i,t} - Y_{i,t}^{(12)} \right)^2, \tag{2.36}$$

where $\mathcal{S}$ denotes the evaluation sample and $|\mathcal{S}|$ is its number of loan-month observations. Lower values indicate better probabilistic accuracy (Brier, 1950).

The Brier score can be decomposed into reliability, resolution, and uncertainty. For binned calibration diagnostics, let predictions be grouped into B bins, with n_b observations in bin b , $n = |\mathcal{S}|$, average predicted probability $\bar{\pi}_b$, observed default rate $\bar{y}_b$, and overall default rate $\bar{y}$. The grouped Murphy decomposition is

$$BS_{\text{grouped}} = \underbrace{\sum_{b=1}^B \frac{n_b}{n} (\bar{\pi}_b - \bar{y}_b)^2}_{\text{Reliability}} - \underbrace{\sum_{b=1}^B \frac{n_b}{n} (\bar{y}_b - \bar{y})^2}_{\text{Resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{Uncertainty}}. \tag{2.37}$$

Reliability measures calibration error. Resolution measures the ability of the model to separate groups with different observed default frequencies. Uncertainty depends only on the base default rate (Murphy, 1973; Dimitriadis et al., 2020). The grouped decomposition is a diagnostic summary; the ungrouped Brier score in Equation 2.36 remains the primary reported score.

Calibration is also evaluated with bin-level observed-versus-predicted default rates:

$$\widehat{\text{Cal}}_b = \left(\frac{1}{n_b} \sum_{(i,t) \in b} \hat{\pi}_{i,t}, \frac{1}{n_b} \sum_{(i,t) \in b} Y_{i,t}^{(12)} \right), \quad b = 1, \dots, B. \quad (2.38)$$

A well-calibrated model should produce bin-level points with small vertical deviations from the 45-degree line; the reliability term in Equation 2.37 summarises these deviations numerically. Calibration is examined overall, by calendar period, and by regime slice. The regime-slice check is central to the thesis because the proposed model is intended to improve probability quality during macro stress and transition periods.

Secondary metrics include log loss, ROC-AUC, and PR-AUC. Log loss is

$$LL = -\frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} \left[Y_{i,t}^{(12)} \log(\hat{\pi}_{i,t}) + (1 - Y_{i,t}^{(12)}) \log(1 - \hat{\pi}_{i,t}) \right]. \quad (2.39)$$

For numerical stability, predicted probabilities are clipped to $[\varepsilon, 1 - \varepsilon]$ before evaluating the logarithms, with $\varepsilon = 10^{-15}$ fixed before testing. ROC-AUC is retained as a ranking measure, while PR-AUC is reported because serious delinquency is rare and precision-recall analysis is informative for imbalanced binary outcomes (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).

Table 2.3: Evaluation metrics and interpretation

Metric	Main question answered	Role in this thesis
Brier score	What is the mean squared error of the predicted PDs?	Primary probability-quality metric.
Reliability diagram	Do predicted PD bands match observed default rates?	Main visual calibration diagnostic.
Brier decomposition	Are changes driven by reliability, resolution, or base-rate uncertainty?	Explains the source of model improvement.
Log loss	Does the model assign high probability to realised outcomes?	Secondary proper scoring rule.
ROC-AUC	Does the model rank defaulters above non-defaulters?	Secondary discrimination metric.
PR-AUC	How well does the model identify rare default events?	Secondary rare-event metric.

The empirical conclusion will be based on the full pattern of evidence rather than one metric alone. The regime-aware model is considered useful if it improves calibration, especially reliability and regime-slice calibration; any deterioration in ROC-AUC or PR-AUC

relative to the baselines is reported explicitly as a discrimination trade-off. This criterion follows the purpose of the model: to produce interpretable and macro-sensitive PD estimates, not merely to maximise a black-box accuracy measure.

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